Kevin David Hayes
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EDUCATION

- Ph.D., Mechanical Engineering, University of Maryland, expected 2027 (Advisor: Dr. Peter Chung)
- M.S., Mechanical Engineering, University of Maryland, 2025
- B.S., Mechanical Engineering, University of Maryland, 2022

RESEARCH EXPERIENCE

- **Computational Research in Science and Technology Lab, University of Maryland**, Graduate Research Assistant Sep 2021 – Present
  Advisor: Dr. Peter Chung.
  - Studied LLM's and VLM applications to chemistry domains
  - Developed new physics viscosity theory for void entrapment in viscous fluids
- **Lawrence Livermore National Laboratory**, Research Intern May 2025 – Sep 2025
  - Trained large language diffusion models at scale.
  - Evaluated diffusion LLMs for scientific reasoning.
- **Department of Defense**, Research Intern May 2024 – Sep 2024
  - Received the Director's Award
- **MEMS and Microfluidics Lab, Fischell Institute for Biomedical Devices, University of Maryland**, Research Assistant Sep 2021 – Dec 2021
  Advisor: Dr. Don DeVoe
  - Planned, repaired, and manually calibrated a retrofitted particle-counting system for condensation growth tube experiments in a BSL-2-compliant lab.

PUBLICATIONS

Conference Papers

- **Kevin David Hayes**, Arka Pal, Haosong Zhang, Tom Goldstein, Micah Goldblum, "Pinocchio: Fast Uncertainty Estimates for Black-Box Language Models," Conference on Language Modeling (COLM), 2026. arXiv:2609.24881.
- Arka Pal, Kwok Chun Au, Louai Zahran, Rahul Krishna Thomas, **Kevin David Hayes**, Micah Goldblum, Adam Block, "Answering At Any Cost: Frontier LLMs Are Consequence-Insensitive," Conference on Neural Information Processing Systems (NeurIPS), 2026. OpenReview.
- **Kevin David Hayes**, Micah Goldblum, Vikash Sehwag, Gowthami Somepalli, Ashwinee Panda, Tom Goldstein, "FineGRAIN: Evaluating Failure Modes of Text-to-Image Models with Vision Language Model Judges," Conference on Neural Information Processing Systems (NeurIPS), 2025 *(Spotlight)*. arXiv:2512.02161.
- **Kevin David Hayes**, Marcus Nelson, Peter Chung, "Machine Learning Techniques for Propellant Void Prediction," JANNAF, 2025.
- **Kevin D. Hayes**, Baily M. Felix, Ryan D. Sochol, "Generative AI-Based Approach for Designing 3D-Printed Microfluidic Devices," 28th International Conference on Miniaturized Systems for Chemistry and Life Sciences (μTAS), 2024.

Journal Articles

- Connor O'Ryan, **Kevin D. Hayes**, Francis G. VanGessel, Ruth M. Doherty, William Wilson, John Fischer, Zois Boukouvalas, Peter W. Chung, "An Automated Approach for Domain-Specific Knowledge Graph Generation – Graph Measures and Characterization," Journal of Chemical Information and Modeling, 65(3), 1243–1257, 2025. doi:10.1021/acs.jcim.4c01904.

Conference Papers Under Review

- Khalid Saifullah, **Kevin David Hayes**, Brian R. Bartoldson, Tom Goldstein, "DriftRetrieval: Learning from Many Negatives with Less Backpropagation," submitted to ICLR 2027.
- **Kevin David Hayes**, Gourab Saha, Brian R. Bartoldson, Tom Goldstein, "Measuring the Attention Span of Long Context Training Data," submitted to ICLR 2027.
- Rifaa Qadri, **Kevin David Hayes**, Brian R. Bartoldson, Bhavya Kailkhura, Arjun Guha, Tom Goldstein, "SynthCode: Not All Synthetic Code Is Worth the Same Tokens," submitted to ICLR 2027.
- **Kevin David Hayes**, Alex Hanson, Allen Tu, Pranav Asthana, Matthias Zwicker, Tom Goldstein, "RIDE-Splat: Reflection Removal for In-the-Wild Driving Environments," preprint, 2026. ridesplat.github.io.

Abstracts and Talks

- **Kevin Hayes**, Peter Chung, "Sentiment and Retrieval-Augmented Generation in Large Language Models for Energetic Materials Discovery," Purdue Energetics Research Center Energetics Talent Showcase Conference, 2024.
- Connor O'Ryan, Frank VanGessel, Zois Boukouvalas, Mark Fuge, Peter Chung, Ian Michel-Tyler, Ruth Doherty, William Wilson, **Kevin Hayes**, "Automated Knowledge Graph Generation from Text for Synthesis of Energetic Materials," Bulletin of the American Physical Society, Vol. 67, 2022.

AWARDS AND HONORS

- Spotlight paper, NeurIPS 2025 (FineGRAIN)
- Director's Award, Chief Engineer of Advanced Systems and Technology, Department of Defense, 2024
- Northrop Grumman Scholarship, 2021

TEACHING

- Teaching Assistant, ENME 332: Heat Transfer, Department of Mechanical Engineering, University of Maryland

SERVICE

- Reviewer for NeurIPS, ICLR, IEEE